

Guided Notes — Introduction to Functions

Worksheet II — Math Summer Camp

Name: _____ Date: _____

Taking notes is a skill.

Here we provide a simple framework to help you get started on learning to take notes. Each concept box has three sections. Fill them in as Leo talks.

DEFINITION ← Copy this exactly. Precision in math depends on exact wording.

INTUITION ← Rephrase it in your own words. If you can't, you don't have it yet.

EXAMPLE ← Write a worked example. This is what you'll look at when homework gets hard.

Hour 1 — What Is a Function?

Function

DEFINITION

← *write this verbatim — precision matters*

INTUITION

← *write this in your own words*

EXAMPLE

← *draw a valid arrow diagram here*

A helpful note-taking tip is to write “Refer to Example 2.2.1 (second week, second lesson, figure 1)” here. Then write the problem and draw the figure in your notebook, and write Example 2.2.1 next to it somewhere you can easily find.

A function f from set A to set B is written:

Four ways to represent a function:

Arrow diagram	Ordered pairs

Table	Graph

What breaks the definition? (Counterexample)

The rule that makes something **not** a function:

Vertical Line Test

DEFINITION

← write this verbatim — precision matters

INTUITION

← write this in your own words

EXAMPLE

← sketch one graph that passes and one that fails

Connection to last week — Set Theory: *(fill in the blanks)*

A function $f : A \rightarrow B$ is actually a special kind of subset of _____.

Special because: each first element appears _____.

Hour 2 — Function Notation, Domain, Codomain, Range

Function Notation $f(x)$

DEFINITION

← write this verbatim — precision matters

INTUITION

← write this in your own words

EXAMPLE

← evaluate $f(3)$ for a function Leo gives

Practice: Given $f(x) = 2x + 1$:

$f(0) =$	$f(3) =$	$f(-2) =$	$f(a) =$

Domain

DEFINITION

← write this verbatim — precision matters

INTUITION

← write this in your own words

EXAMPLE

← identify the domain from a table or graph

Codomain

DEFINITION

← write this verbatim — precision matters

INTUITION

← write this in your own words

EXAMPLE

← identify the codomain from an arrow diagram

Range (Image)

DEFINITION

← write this verbatim — precision matters

INTUITION

← write this in your own words

EXAMPLE

← list the range of $f(x) = 2x + 1$ when domain = $\{0, 1, 2\}$

Set Theory connection: (fill in the blanks)

The range is always a _____ of the codomain.

In set notation: $R(f) \subseteq$ _____.

Practice — for each function below, identify domain, codomain, and range:

Function	Domain	Codomain	Range
$f : \{1, 2, 3\} \rightarrow \{A, B, C, D\},$ $f(1) = A, f(2) = A, f(3) = B$			
$g : \mathbb{R} \rightarrow \mathbb{R}, \quad g(x) = x^2$			
$h : \{\text{cats}, \text{dogs}\} \rightarrow \{0, 1, 2\},$ $h(\text{cats}) = 2, \quad h(\text{dogs}) = 2$			

Hour 3 — Injective, Surjective, Bijective

Injective (one-to-one)

DEFINITION

← write this verbatim — precision matters

INTUITION

← write this in your own words

EXAMPLE

← draw an injective arrow diagram

Surjective (onto)

DEFINITION

← write this verbatim — precision matters

INTUITION

← write this in your own words

EXAMPLE

← draw a surjective arrow diagram

Bijjective

DEFINITION

← write this verbatim — precision matters

INTUITION

← write this in your own words

EXAMPLE

← draw a bijective arrow diagram

Quick check — label each diagram below as injective, surjective, bijective, or none:

Diagram 1 (Leo draws)	Diagram 2 (Leo draws)	Diagram 3 (Leo draws)
Label: _____	Label: _____	Label: _____

Cantor connection: (fill in the blanks)

A bijection between two sets means they have the same _____.

Cantor used this idea to compare the sizes of _____ sets.

Challenge Problems

C1. Let $A = \{1, 2, 3\}$ and $B = \{a, b, c, d\}$.

- (a) Can a function $f : A \rightarrow B$ be surjective? Explain.
- (b) Can a function $f : A \rightarrow B$ be injective? Construct one.

C2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$.

- (a) Is f injective? Give a counterexample or a proof.
- (b) Is f surjective? Justify.
